



Development of cellulose-based superabsorbent polymers: a review

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Abstract Superabsorbent polymers (SAPs) are three-dimensional crosslinked hydrophilic polymers, that can absorb and retain liquids up to hundreds of times their weight. These polymers have diverse applications across various fields, including agriculture, biomedicine, separation technologies, and wastewater treatment. Among these cellulose-based SAPs are prominent due to their biodegradable nature, sustainability, biocompatibility, cost-effectiveness, and natural abundance. The development of SAPs has evolved significantly since 1961, marked by advancements in polymerization techniques and their integration into numerous aspects of daily life. This review provides a comprehensive analysis of recent advancements in the field of cellulose-based superabsorbents, focusing on their polymerization methods, source of cellulose, and diverse applications. Furthermore, it highlights the mechanisms by which different forms of cellulose can enhance liquid absorption capacity and kinetics across various applications. The findings underscore the importance of cellulose-derived SAPs in promoting environmentally sustainable practices while addressing the growing demand for effective

water retention solutions in agricultural and industrial contexts.

Keywords Absorbance capacity · Biodegradation · Cellulose · Hydrogels · Polymerization technique · Renewable · Superabsorbent Polymer

Introduction

Superabsorbent polymers (SAPs) are extensively crosslinked hydrogels with a remarkable ability to absorb and retain large volumes of liquids, often exceeding their weight by several hundred times. Hydrogels are formed by cross-linking hydrophilic monomers, creating a 3D network that can absorb large volumes of liquid (Behera and Mahanwar 2020). Their diverse applications have revolutionized industries, ranging from hygiene products to agriculture, controlled drug release, artificial muscles, horticulture, tissue engineering, and water management (Suo et al. 2007a, b), and have earned them a prominent place in modern materials. However, conventional SAPs primarily rely on petroleum-based monomers, raising concerns about their non-biodegradability and environmental footprint (Fu et al. 2022). Their accumulation in landfills contributes to plastic pollution, while their production relies on depleting resources.

In 1961, Russell (Ma and Wen 2020) produced the first superabsorbent polymer (SAP) by grafting acrylonitrile on starch. Fanta et al. (1967) later found that

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SAPs made from starch derivatives exhibited great water absorption capacity and were capable of retaining water under pressure. In the 1970s, the Northern Regional Laboratory of the United States Department of Agriculture developed the first-ever SAP using starch-graft-hydrolyzed acrylonitrile polymers (Athawale et al. 2001). The commercial SAP production for female napkins began in Japan in 1978. In the early 1980s, both France and Germany were using SAP materials in child diapers (Ramazani-Harandi et al. 2006). By the 1990s, global SAP production resins had reached one million tons. Prominent SAP manufacturers include Stockhausen, Hoechst, Amcol (Chemdal), Sumitomo, SNF Floerger Companies Colon, and Nalco (Kazanskii and Dubrovskii 2005).

Dry polymeric material classified as SAP must exhibit a high absorption capacity, typically exceeding 20 times its weight in water, saline, or physiological solutions. This ability to imbibe significant quantities of liquid distinguishes SAPs from conventional absorbent materials. Upon absorbing liquid, the resulting hydrogel formed by the SAP should retain its original shape (Oladosu et al. 2022). This characteristic prevents undesirable leakage or dripping of the absorbed solution and is crucial for applications where spatial confinement of the SAP is essential. The swollen hydrogel possesses adequate physical integrity to resist deformation and flow. This property ensures the retention of the absorbed liquid within the SAP structure, even under moderate pressure or stress. Such resistance to flow is critical for applications like hygiene products and soil management, where liquid containment is paramount. The SAP possesses the capability to retain the absorbed solution under a certain pressure. This characteristic prevents premature release of the liquid, particularly when the material is subjected to compression or external forces. Pressure retention is relevant for applications such as wound dressings and agricultural settings where optimal water management is required (Xie et al. 2012).

These fundamental requirements, alongside other parameters like rate of absorption, desorption properties, biodegradability, and toxicity, collectively determine the suitability of a polymeric material as an SAP for a specific application. Identifying an appropriate SAP entails selecting a material possessing the optimal combination of these properties to effectively fulfill the desired function.

SAPs are categorized into six types based on their raw materials: starch-based, cellulose-based, protein-based, synthetic polymers, chitosan-based, and composites or blends (Ma and Wen 2020). Superabsorbent polymers typically employ acrylic acid, its salts, and acrylamide as the primary monomeric components. The exceptional water-holding capacity of these materials stems largely from the presence of polar functional groups within the polymer chains, specifically amino ($-NH_2$), carboxyl ($-COOH$), and hydroxy ($-OH$) groups. These groups readily interact with water molecules via hydrogen bonding, facilitating extensive hydration and swelling of the polymer network (Chang et al. 2010).

Organic superabsorbent materials like chitosan, starch, carrageenan, and cellulose originate from renewable resources like microorganisms and plants, making them a more environment-friendly option than conventional manufactured polymers. They are biodegradable, have minimal environmental impact, and have biocompatible and non-toxic properties, positioning them as suitable alternatives for sustainable superabsorbent development (Masuda and Ueda 2014).

Synthetic SAPs are popular but non-biodegradable; hence, they accumulate and impact the environment, and contribute to soil and water pollution. Researchers are exploring novel, sustainable, and biodegradable materials as they are eco-friendly. Biodegradable materials have minimum impact on the environment. Sodium lignosulfonate and cellulose-based materials have shown significant water absorption capacity and water retention. Innovative products from these materials show potential applications in various fields (Wang et al. 2018; Kassem et al. 2020).

Cellulose, nature's abundant and renewable resource, provides a promising alternative to the synthesis of eco-friendly, less toxic, biodegradable superabsorbents. It is a linear polysaccharide composed of glucose units connected by β -1,4-glycosidic bonds. Cellulose comes mostly from the cell walls of plants and is an excessively available, renewable, and biodegradable material. Along with the existence of hydroxy groups, cellulose is hydrophilic and provides opportunities for chemical modification, enabling changes in absorption properties and biodegradability, which makes it a perfect precursor for superabsorbent polymers. Cellulose is readily available and cost-effective compared to petroleum-based

monomers, potentially leading to more affordable SAPs (Etminani-Isfahani et al. 2020). Cellulose and carboxymethyl cellulose have shown significant biodegradable and sustainable characteristics. The use of cellulose-based superabsorbent hydrogels is escalating the attention of agriculture, medicine, environmental management, and construction. Additionally, they are biodegradable, show high water-absorption capacity, and are versatile. Their use in agriculture has appeared to enhance soil fertility and water retention (Alam and Christopher 2018; Wei et al. 2021). The biodegradable nature of cellulose-based hydrogels has been utilized in various applications. Cellulose-based hydrogels are hemostatic agents and control immense bleeding (Mahmoodzadeh et al. 2021). Cellulose-based hydrogels have shown potential use in environmental management. These are used to minimize pollution (Jeddi et al.). Superhydrophobic aerogels from cellulose-based materials absorb oil efficiently and are used in construction and engineering (Mi et al. 2018).

Review

Sources of cellulose

There are two main sources for the preparation of SAP.

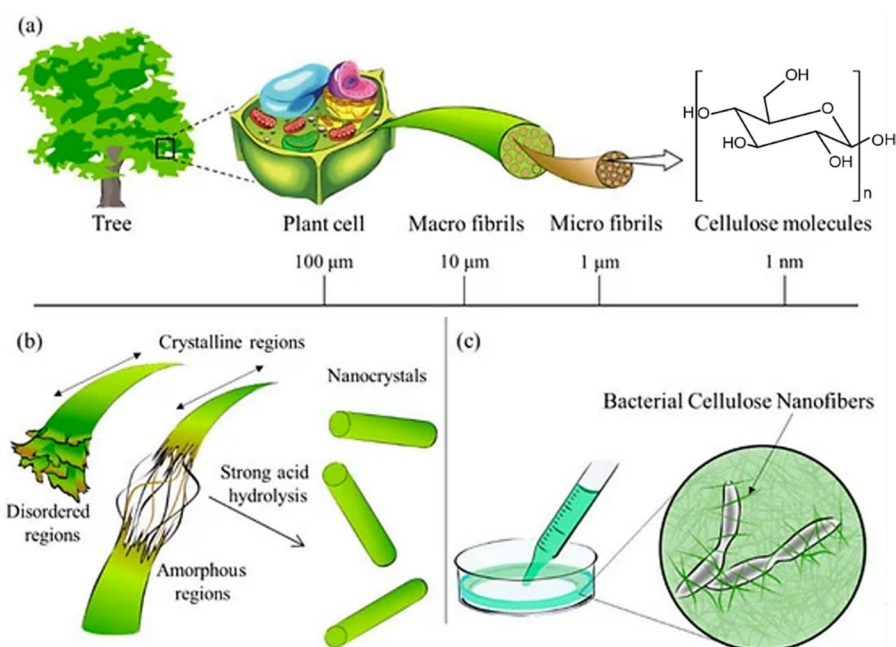
1. Natural sources
2. Cellulose derivatives

Natural sources of cellulose

Native cellulose, derived from various natural plants, possesses inherent advantages that make it a valuable component in the synthesis of superabsorbent hydrogels. One of its notable attributes is biodegradability, ensuring environment-friendly end-products. The ability of native cellulose to undergo repetition usage adds to its appeal, making it a sustainable choice for superabsorbent polymer applications. Figure 1 represents the hierarchical structure of cellulose and its transformation into nanocellulose and bio-nanocellulose.

Examples of plants providing native cellulose include:

Fig. 1 Cellulose contained in plants or trees has a hierarchical structure from the meter to the nanometer scale, as shown in (a). A schematic diagram of the reaction between cellulose and strong acid to obtain Nanocellulose is shown in (b). Bio-nanocellulose (c) (Miyashiro et al. 2020)



Wheat straw

Wheat straw is a valuable byproduct of grain harvests. It contains high levels of hemicellulose and natural cellulose, and the lignin. However, this resource is often underutilized, with most of it being incinerated, leading to environmental pollution and wastage of natural resources. Wheat straw can be chemically changed and utilized as a skeleton material for grafting monomers like AA and AM to generate a superabsorbent composite. Zuohao et al. developed SAP by the graft copolymerization of acrylic acid onto wheat straws with a water absorption capacity of 133.76 g/g in distilled water and about 33.83 g/g in 0.9 wt.% NaCl solution (Ma et al. 2011). Wan et al. prepared a novel hydrogel with maleic anhydride-modified wheat straw in an aqueous solution, it took only 13.1 min for the wheat straw composite superabsorbent to absorb water to 63 percent of its equilibrium absorbency (781 g/g) (Wan et al. 2013). The utilization of wheat straw showcased not only the feasibility of using agricultural waste but also demonstrated the potential for improving water absorbency in hydrogel structures.

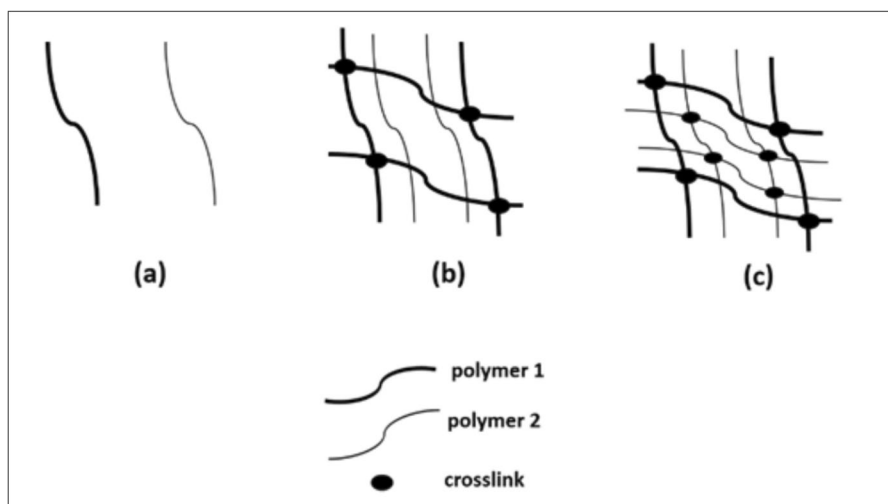
Cotton stalks

Cotton, a widely cultivated crop, is another valuable source of native cellulose. Its cellulose content has been explored for superabsorbent polymer synthesis, capitalizing on the structural integrity and water absorbency characteristics inherent in cotton stalks. Cotton stalks, a byproduct of cotton cultivation, pose

both an environmental challenge and an opportunity. As cotton production increases, so does the volume of these stalks, which are often discarded or burned in the field (El-Hendawy et al. 2008). However, these seemingly insignificant remnants hold great potential. Rich in cellulose, they can be harnessed for various purposes without competing with food or animal feed production. Efforts are underway to transform cotton stalks into valuable resources. By converting them into biofuels, particle boards, and other industrial applications, we can address both environmental concerns and resource scarcity. Additionally, research into biodegradable superabsorbent materials from renewable sources is gaining prominence (Nahil and Williams 2012). Ultimately, repurposing this agricultural waste can offer a sustainable and eco-friendly solution for disposal while advancing scientific innovation. The versatility of cotton-derived cellulose contributes to the adaptability of these hydrogels for diverse applications. Figure 2 shows the crosslinking between two different types of polymers.

Sawut et al. prepared a new biodegradable SAP made using cellulose from the cotton stalk and acrylic acid was developed. Its absorbency and properties were tested under various conditions like temperature, exposure time, and concentration. The resulting material showed good swelling ability and water retention, making it promising for agricultural applications due to its eco-friendliness and affordability (Sawut et al. 2014). The study by Wen et al. (Wen et al. 2016) presented a novel method for synthesizing a semi-interpenetrating polymer network (semi-IPN)

Fig. 2 **a** Two different types of polymers. **b** Semi-interpenetrating **c** Interpenetrating polymer network (Khalil 2020)



to release the nitrogen fertilizer slowly with water absorption using microwave irradiation. The semi-IPN structure and the bentonite clay help the fertilizer absorb and retain water, which reduces nutrient leaching and improves plant growth. The fertilizer's water absorbency was highest when it was made with 2.5 g of cotton stalk, 1.5 g of PVP, and 0.5 g of bentonite, and irradiated for 3 min at 160 watts. The fertilizer also released about 60% of its nutrients over 30 days, which is a slow and steady release that is ideal for plant growth.

Flax and mulberry branches

Beyond cereal crops, other plants like flax and mulberry branches have been investigated for their cellulose content. These plant-derived cellulose sources play a prominent role in the structural framework of plant cells, contributing to the stiffness and water absorption capacity needed for effective superabsorbent hydrogels. In recent years, there has been growing interest in efficiently utilizing agro-industrial residues (Wang et al. 2008). Mulberry, an economically significant plant, is widely cultivated across Asia, North and South America, Europe, and Africa (Wasano et al. 2009). While mulberry is primarily grown for its leaves to facilitate silkworm rearing, the piles of mulberry twigs are sometimes discarded as agricultural waste or used as firewood (Liu et al. 2010). However, recent advancements have led to progress in this field. For instance, cellulose whiskers have been extracted from mulberry bark for potential reinforcement applications (Li et al. 2009), and pectin has been obtained from both epidermis-bearing and epidermis-free mulberry bark. Furthermore, natural fibers have been isolated from the endothelium of mulberry branches (García et al. 2012). Considering that mulberry branches mainly consist of lignin, hemicelluloses, and cellulose; they can serve as valuable supporting material. By grafting monomers like acrylic acid (AA) and acrylamide (AM) onto mulberry branches, superabsorbent composites can be created for potential applications in agriculture and horticulture. In the study by Liang et al. (Zhang et al. 2014) Mulberry branches were used to create a new eco-friendly superabsorbent that can hold over 570 times its weight in water. This material shows promise for agriculture because of its high water retention and ability to slowly release nutrients like urea.

Bacterial cellulose (BC)

In recent times, plant cellulose has garnered attention as a promising substrate for synthesizing superabsorbents. Its structural similarity to starch (both being polysaccharides with glucose units connected by different glycosidic linkages) and numerous advantages, including low cost, widespread availability, and renewability makes it an appealing choice (Avcioglu 2022). However, when considering factors like purity and durability, cellulose from plants still falls short of being a perfect substrate for superabsorbent production. To address this limitation, microbial cellulose, specifically bacterial cellulose, emerges as a more suitable alternative (Rastogi et al. 2020).

Bacterial cellulose is a useful extracellular biopolymer synthesized by various bacteria, including *Sarcinia*, *Lactobacillus*, *Acetobacter*, *Achromobacter*, *Sarcinia*, *Lactobacillus*, *Komagataeibacter*, and *Agrobacterium* (Lu et al. 2020), *Bacillus*, *Azotobacter* (Soemphol et al. 2018). Compared to plant cellulose, BC does not contain lignin, hemicellulose, pectin, arabinose, or other plant-derived contaminants. As a result, BC can be extracted directly from culture media with no requirement for purifying steps (Abdelraof et al. 2019). BC boasts outstanding physical, chemical, and mechanical properties, including quality, exceptional crystallinity, transparency, permeability, water retaining capacity, an ultrafine nanoscale fiber network (Carreira et al. 2011), tensile strength, a high degree of polymerization, and a large surface area. Moreover, BC is chemically stable, exhibits proton conductivity, and is both biocompatible and biodegradable (Abdelraof et al. 2019). It has become an indispensable nanomaterial in various industrial applications due to its renewable nature.

Bacterial cellulose, as a polysaccharide, possesses remarkable properties, as highlighted by research studies (Du et al. 2018). These studies have demonstrated the feasibility of utilizing BC as a substrate for superabsorbent synthesis (Abeer et al. 2014). Despite its potential applications, the costly price of BC feedstock remains an important hurdle for its industrialization. Currently, BC is more expensive than plant cellulose, impacting the market competitiveness of BC-based superabsorbents. Novel SAP was synthesized by Luo et al. from bacterial cellulose on bentonite inorganic gel (BIG). The synthesis process was optimized by taking into account temperature, gelling

agent content, and sodium agent type. The polymerization method took into account temperature, monomer-to-substrate ratio, initiator and crosslinker content, and reaction temperature. The composite that was produced demonstrated good water and salt absorption as well as water retention (Luo et al. 2019). A new process was developed by Chaiyasat et al. to produce superabsorbent hydrogels from bacterial cellulose. The BC was modified through carboxymethylation and then functionalized with glycidyl methacrylate before being crosslinked with polyethylene glycol diacrylate (PEGDA). Increasing the amount of PEGDA resulted in a denser network with fewer capillary spaces. These hydrogels exhibited maximum water retention of 125 g/g in distilled water and 29 g/g in saline, with pore sizes ranging from 350 to 600 μm , allowing for rapid mass penetration. They achieved 80% of their maximum water absorption capacity within 30 min (Chaiyasat et al. 2019). Pandey et al.'s study evaluated how solubilized and dispersed BC affects the properties and drug release profiles of hydrogels. The results indicated that hydrogels made with solubilized BC had higher swelling ratios, porosity, drug loading efficiency, and

drug release compared to those made with dispersed BC. These findings suggest that hydrogels using solubilized BC are superior and hold promise for further exploration in oral drug delivery applications (Pandey et al. 2013).

While the major sources of cellulose have been discussed in detail, several other natural sources also contribute to its diversity in properties and applications. To provide a comprehensive overview of diverse natural cellulose sources and their potential applications in superabsorbent development, a detailed summary is presented in Table 1.

Cellulose derivatives

Cellulose, a versatile biopolymer, has the remarkable ability to form gels. However, traditional methods involving special solvents like alkali-urea (Yang et al. 2020), DMAc-LiAc, or ionic liquids are often used to produce cellulose-based hydrogels (Mischnick et al. 2010). These solvents present challenges, prompting researchers to explore alternative approaches. One promising avenue is the development of cellulose-based hydrogels through inter- and intra-molecular

Table 1 Sources of cellulose with key characteristics and applications

Cellulose source	Characteristics	Applications
Water Hyacinth (Fibers)	Abundant, biodegradable, low thermal stability, ~80–150 g/g water absorption	Wound dressing, agriculture, flame-retardant coating (Tanpichai et al. 2022)
Wood pulp (Fibers)	High structural strength, abundant availability, Biodegradable, ~30–70 g/g water absorption	Agriculture, sanitary products, biomedical, environmental remediation, industrial applications (Alam and Christopher 2018; Cao et al. 2022)
Algae-derived cellulose (Microfibers)	Aquatic, biodegradable, eco-friendly, renewable, ~100–200 g/g, water absorption	Agriculture, wastewater treatment, healthcare (Dodangeh et al. 2024)
Rice Husk (Fibers)	Rich in cellulose, low-cost, biodegradable, ~50–100 g/g water absorption	Water reservoirs, agriculture, controlled water release systems (Kenawy et al. 2021)
Banana fiber waste (Fibers)	Low cost, ecological, biodegradable, ~4–10 g/g water absorption	Used in agriculture for soil moisture retention, Effective in dye removal from wastewater (Ahmed Yassine et al. 2024)
Sugarcane Bagasse (Fibers)	Abundant byproduct, biodegradable, ~150–300 g/g water absorption	Agricultural, oil industry, eco-friendly SAPs (Riswati et al. 2021)
Hemp (Fibers, microfiber)	High cellulose content, durable, biodegradable, ~60–120 g/g water absorption	Personal care products, water management (Grace Kakonke 2019; Sun et al. 2023)
Jute (Fibers)	Low cost, coarse fibers, durable, rich in cellulose, ~50–100 g/g water absorption	Personal care products, Agriculture (Sharma et al. 2014)
Soybean dregs (Fibers)	Rich in cellulose, thermal stability, ~200–300 g/g water absorption	Agricultural, forestry, Biomedical, horticulture (Fan et al. 2018)
Oil Palm empty fruit bunch (Fibers)	Fibrous, high strength, thermal stability, ~100–200 g/g water absorption	Agriculture, horticulture, sanitary products (Salleh et al. 2019)

crosslinking in colloidal and aqueous solutions of cellulose derivatives. By substituting hydroxy groups with hydroxyethyl, carboxymethyl or other functional groups on the backbone of the cellulose, properties such as increased water swelling capacity can be achieved (Fekete et al. 2014a). The qualities of cellulose derivatives are influenced by the specific kind of substituent, the molecular weight, the degree of substitution, and substituent distribution. Cellulose derivatives exhibit the capability to create primarily physical hydrogels or crosslinked chemical hydrogels, based on the kind of functional groups involved. In physically associated hydrogels, the cellulose derivative chains aggregate through hydrogen bonds, hydrophobic forces, or ionic interactions. For the chemical crosslinking of cellulose derivatives, different agents, and catalysts come into play (Ciolacu et al. 2020). Table 2 describes the advantages and disadvantages of different cellulose derivatives along with their applications.

Recent studies have synthesized cellulose derivative-based hydrogels using various techniques (Yang et al. 2011). While carboxymethyl cellulose remains a focal point, other derivatives, including alkyl and hydroxyalkyl variants (Uraki et al. 2006), also hold promise for applications in fields including drug delivery, industrial materials, and tissue engineering. A summary of different SAP applications of cellulose derivatives is shown in Table 2.

Carboxymethyl cellulose (CMC)

CMC is a prominent cellulose derivative as shown in Fig. 3. It is used for making cellulosic superabsorbent materials. It stands out due to its controlled properties and cost-effectiveness compared to other materials with similar solution properties. The process involves the treatment of alkali cellulose with monochloroacetic acid to produce sodium carboxymethylcellulose (Nevell and Zeronian 1985). This white or yellowish, odorless derivative is highly hygroscopic. When partially carboxymethylated cotton is used, it exhibits increased moisture absorbance, enhanced absorption, and high swellability. These properties are partly attributed to the hydrophilicity of the carboxyl groups. Additionally, the bulky carboxyl groups play a role in opening up the cellulose structure, allowing water to diffuse into microscopic spaces. Alkali treatment ensures that carboxymethyl

substituents are integrated into previously inaccessible areas, preventing the structure from collapsing after drying. Consequently, these modified regions become available for liquid penetration, which is crucial for developing superabsorbent CMC polymers (Young 2002). The carboxyl functional groups enhance the water absorption capacity of the resulting hydrogel, making CMC a versatile choice for applications in agriculture, healthcare, and wastewater treatment. Djafari et al. developed Superabsorbent Polymers with a swelling capacity of 200 gr/gr and ionic sensitivity ranging from 0.73–0.98. These figures decreased with increased citric acid and TEMPO-mediated cellulose nanofibers (TCNFs) content. The SAPs are biodegradable and suitable for hygiene and horticulture applications (Djafari Petroudy et al. 2018). Ko et al. produced superabsorbent polymers of carboxymethyl cellulose-graft-itaconic acid by using N, N'-methylene bisacrylamide, and potassium persulfate. A hydrogel with a swelling capacity of 74 g/g in DI water and 18 g/g in 0.9 wt% NaCl solution was produced as a result of the optimized reaction variables. Diffusion exponents (n) between 0.73 and 0.93 were seen in the swelling kinetics when the Fickian diffusion model and Schott's pseudo-second-order kinetics model were used to examine the data. This indicates a considerable chain relaxation behavior throughout absorption (Ko et al. 2018). The gamma irradiation method was used by Fekete et al. to develop hydrogel using CMC. Partially substituting starch for CMC improved the gel fraction along with water absorption, but too much starch negatively affected gelation. The best hydrogels had a gel fraction of around 55% and a water absorption of about 350 water/g of gel when they contained 30% starch. These gels, which had a heterogeneous structure with starch granules dispersed in the CMC matrix, were sensitive to ionic strength but less so than pure CMC gels (Fekete et al. 2017). Recently A dual crosslinking strategy was used to create interpenetrating polymer network (IPN) hydrogels based on CMC by Jeong et al. (Jeong et al. 2020) with superabsorbent properties and high mechanical strength. The CMC/PAM IPN hydrogels, characterized by various techniques, showed enhanced mechanical strength due to double crosslinking. In comparison to a CMC gel, the tensile length and compression strengths were up to 2.6 and 4.5 times higher, respectively. These hydrogels also exhibited superior superabsorbent properties

Table 2 Cellulose derivatives with their advantages and disadvantages

Cellulose Derivatives	Applications	Advantages	Disadvantages	References
Carboxymethyl Cellulose (CMC)	Wound dressings, drug delivery Agriculture: soil moisture retention, seed germination Wastewater treatment: removal of heavy metals	High water absorbency, Biocompatible and biodegradable, Tunable properties, Non-toxic	Can be mechanically weak, Susceptible to enzymatic degradation (depending on application)	Tang et al. (2014), Young (2002), Djafari Petroudy et al. (2018), Jeong et al. (2020)
Methyl Cellulose (MC)	Controlled release of drugs (e.g., tablets) Agriculture (fertilizer release)	Good water absorbency, Film-forming properties, Cost-effective	Lower water absorbency compared to CMC, less mechanically strong	Fekete et al. (2014b), Eyholzer et al. (2011), Niemezyk-Soczynska et al. (2019), DEWAN (2023)
Hydroxyethyl Cellulose (HEC)	Smart materials, controlled release systems, sensors Thickening agents in various products	Responsive to external stimuli (e.g., temperature, pH), Water soluble, Film-forming properties	Lower water absorbency compared to CMC, Requires specific conditions for cryogenic treatment	Adair et al. (2017), Abdel-Halim (2014), Li et al. (2014), Wang et al. (2015)
Hydroxypropyl Methyl Cellulose (HPMC)	Controlled drug release (e.g., tablets, capsules)	Wide range of viscosity grades, Extended drug release profiles, Biocompatible and non-toxic	It can be expensive and May require additional processing steps	Chen and Chen (2019), Pygall et al. (xxxx) Viridén et al. (xxxx)
Cellulose Acetate (CA)	Drug carrier systems (membranes, nanoparticles)	Insoluble in water, Tunable properties for specific drug release	May require harsh chemicals for cross-linking Potential biocompatibility concerns depending on modification	Chen and Chen (2019), Filho et al. (2008), Tingaut et al. (2010), Das et al. (2014)

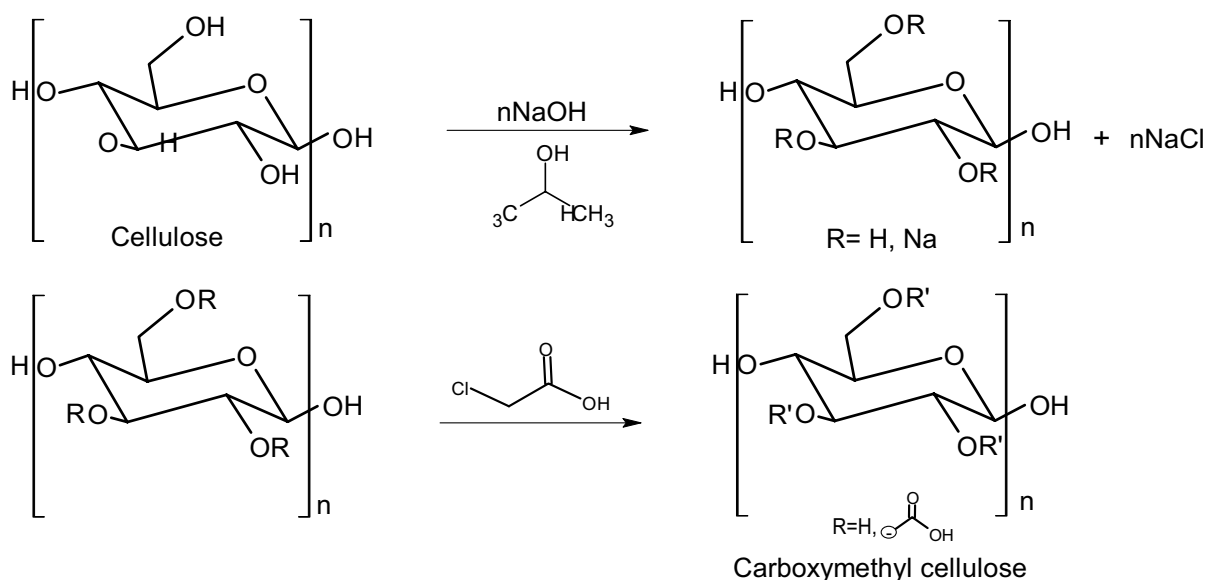


Fig. 3 Synthesis of Carboxymethyl Cellulose from cellulose

and showed no biotoxicity against in vitro animal cells, making them potential biomaterials.

Hydroxyethyl cellulose (HEC)

Hydroxyethyl cellulose (HEC) a type of cellulose ether, is commonly used in preparing superabsorbent cellulose products. It is synthesized in the presence of sodium hydroxide by reacting ethylene oxide with cellulose (Li et al. 2014) as shown in Fig. 4. The sodium hydroxide is not completely consumed during the reaction. An intermediate compound called alkali cellulose forms and reacts with ethylene oxide, it regenerates sodium hydroxide and produces

hydroxyethyl cellulose. The attached hydroxyethyl group contains a free hydroxy group, allowing it to react further with ethylene oxide and add additional hydroxyethyl groups (Abdel-Halim 2014). This process can be repeated several times, leaving a chain of polyoxyethylene connected to the cellulose (Figs. 5, 6).

The molar substitution (MS) of HEC is typically much more than the degree of substitution (DS). The minimal MS value needed for the solubility in water is 1.0, corresponding to a DS of approximately 0.4 (McMullen et al. 2022). Due to the non-uniform substitution of glucose residues, a single cellulose chain may have both water-soluble HEC segments

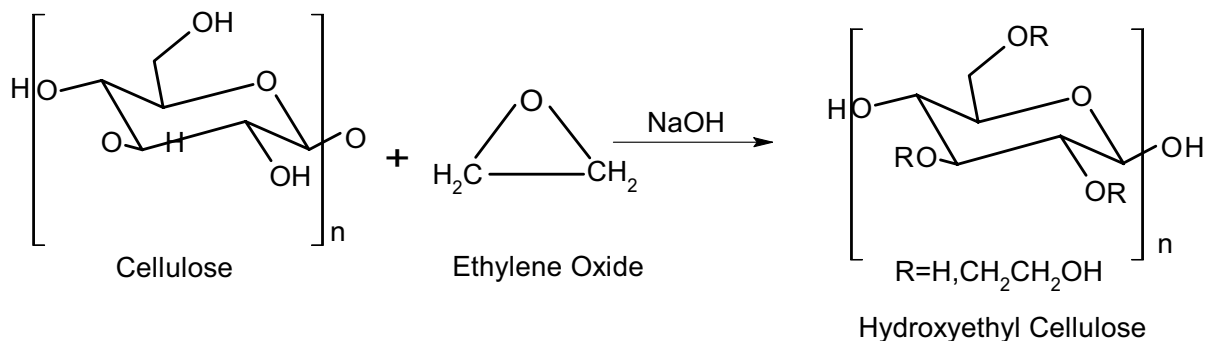


Fig. 4 General reaction for the conversion of cellulose in HEC

Fig. 5 The gel formation process of Methyl Cellulose (DEWAN 2023)

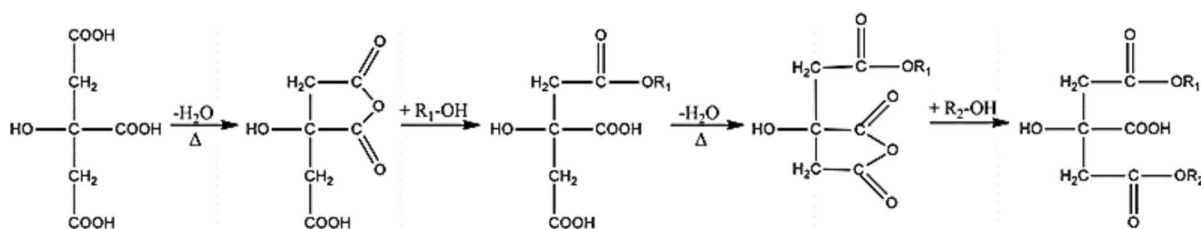
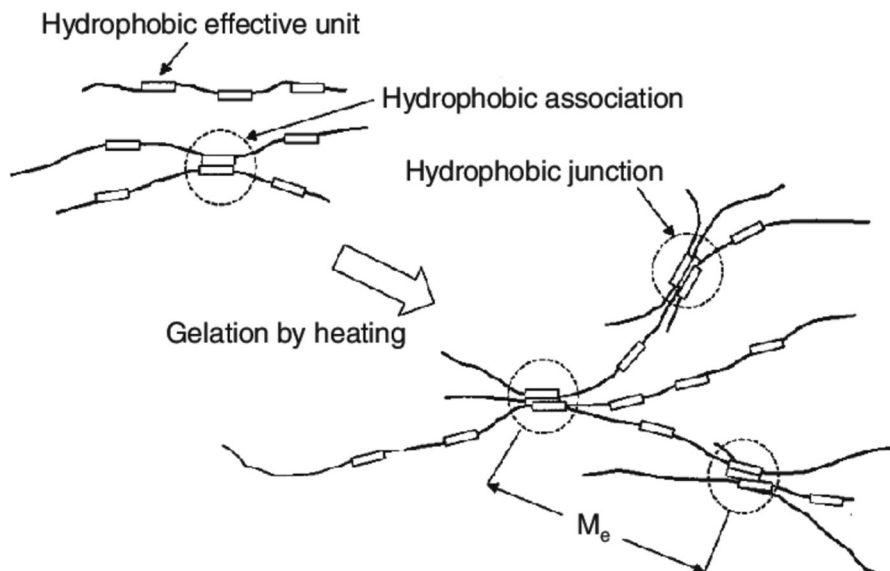


Fig. 6 Citric Acid (CA) based crosslinking of MC (Bonetti et al. 2023)

and unsubstituted cellulose segments. Water competes with cellulose for the reagent, resulting in the formation of byproduct glycols. HEC's higher hydrophilicity is related to its lengthy side chains, which prevent the cellulose from forming securely bound structures, similar to what occurs with CMC (Wang et al. 2015). HEC-modified hydrogels demonstrate improved water absorption and mechanical properties, making them compatible with applications in personal care products and medical devices. Baldino et al. (Baldino et al. 2020) explored an eco-friendly alternative for superabsorbent polymers by using CMC and HEC hydrogels. These hydrogels, dried with a supercritical process, showed exceptional water absorption when the solvent exchange began with 50% ethanol, exceeding other similar methods. This suggests a promising eco-friendly approach for superabsorbent materials. Fawal et al. (El Fawal et al. 2018) combined HEC with tungsten

oxide to create hydrogel membranes for wound dressings. These membranes showed promising properties, both anti-inflammatory and antibacterial properties, making them safe for human cells and potentially effective against common wound bacteria like *Salmonella* and *Pseudomonas*. This suggests their potential as a promising material for wound dressings. A new polymeric hydrogel was created by El-Sayed et al. (El-Sayed et al. 2020) by grafting hydroxyethyl methacrylate onto hydroxyethyl cellulose, with *N,N'*-Methylenebisacrylamide as a crosslinker. The thermal stability of the hydrogels improved as the monomer ratio increased. Cytotoxicity tests on RPE-1 normal cell lines, HCT-116, MCF-7, HepG2, and cancer cells showed that the hydrogels were preferentially damaging to the cancer cells in a dose-dependent way but less toxic to the normal cells. This suggests potential use in anti-cancer drug formulation.

Methylcellulose (MC)

Methyl cellulose, obtained through the methylation of cellulose, is characterized by its thermal gelation properties. This cellulose derivative undergoes a reversible gelation process due to temperature changes, making it suitable for controlled-release applications (DEWAN 2023) as shown in Table 3. MC-modified hydrogels show promise in the biomedical field such as scaffolds for bone, cartilage, muscles, and skin, particularly for synthesizing *in situ* hydrogels for controlled drug delivery systems, where the gelation behavior can be harnessed for the sustained release of pharmaceutical agents. Due to the gel-formation behavior and biodegradability of MC, it is widely used in tissue engineering (Niemczyk-Soczynska et al. 2019).

Eyholzer et al. (Eyholzer et al. 2011) synthesized the bio-composite superabsorbent hydrogels to replace the natural human nucleus pulposus (NP) in intervertebral discs in the presence of carboxymethylated, nanofibrillated cellulose powder. Various artificial and natural superabsorbent hydrogels have effectively employed cellulose microfibers, also known as nanowhiskers or nanofibers, as reinforcing fillers. MC can help form cohesive films within SAP structures, enhancing their mechanical strength and preventing disintegration. It is generally considered safe for human use, making it suitable for applications in personal care and biomedical products. Moreover, MC can be combined with other polymers, such as polyacrylamide or polyvinyl alcohol, to create SAPs with enhanced properties, such as increased water absorption capacity or improved gel strength. These properties make methyl cellulose a versatile choice for a variety of applications. Table 4 presents the schematic reaction of citric acid-based crosslinking of methylcellulose.

Hydroxypropyl methylcellulose (HPMC)

Hydroxypropyl methylcellulose is an ether derivative with hydroxypropyl and methyl substituents. HPMC is recognized for its film-forming and gelling properties, contributing to its use in superabsorbent hydrogel formulations (Pygall et al. 2010). HPMC-modified hydrogels exhibit improved mechanical strength and durability, making them suitable for applications in agriculture and as wound dressings. The

biocompatible and non-carcinogenic material HPMC is used in many products, such as food items, paper, textiles, and pharmaceuticals (Viridén et al. 2011). Cellulose reacts with epoxy, propane, and chloromethane to form modified cellulose or HPMC. HPMC's high content of hydroxypropyl groups contributes to its increased water solubility. In terms of biomaterials, HPMC's water solubility makes sense for its use in creating a superabsorbent hydrogel.

Acetylated cellulose

Acetylated cellulose results from the acetylation of cellulose, introducing acetyl groups to the cellulose structure. This modification enhances the solubility of cellulose in various solvents and broadens its applicability in hydrogel synthesis (Tingaut et al. 2010). Acetylated cellulose derivatives have been explored for their potential in developing environmentally responsive hydrogels, demonstrating applications in controlled drug delivery and smart materials. Archana et al. reported a detailed procedure for the development of acetylated cellulose by using rice husk using eco-friendly conditions (Das et al. 2014). Similarly, Filho et al. prepared and characterized acetylated cellulose from recycled newspapers (Filho et al. 2008). Recycled polymeric catalysts can also be used for the acetylation of cellulose, as studied by Luo et al. (Luo and Sun 2006). Another process was developed by Biswas et al. in which agricultural residues or byproducts such as corn fiber, rice hulls, and wheat straw were used to obtain industrial-grade cellulose acetate along with the removal of hemi-cellulosic sugars. Senna et al. fabricated hydrogels through Esterification and crosslinking with EDTA dianhydride by utilizing acetate cellulose (Senna et al. 2014). The results have shown that the hydrogel with roughly 0.36 EDTA molecules per repetition unit of acetate cellulose had the maximum absorbency. The hydrogels were made with varied levels of crosslinking. Aside from the typical etherification product, it has also been reported that hydrazide or aldehyde functionalized (Patenaude and Hoare 2012) products of cellulose have a role in the construction of superabsorbent hydrogels based on cellulose. For example, in the presence of epichlorohydrin, quaternized cellulose (Chang et al. 2011) was cross-linked with carboxymethyl cellulose in an aqueous solution of NaOH (ECH).

Table 3 Comparison of different cellulose-based SAPs

Type of SAP	Absorption capacity (g/g gel)	Experimental Conditions	Absorption Mechanism	References
Natural Cellulose-based SAP	~ 650	Cross-linked with epichlorohydrin, eco-friendly synthesis	The polymer's hydrophilicity allows it to swell significantly in water due to extensive hydrogen bonding and ionic interactions	Alam and Christopher (2018)
Carboxymethyl Cellulose (CMC)	~ 750	Grafted with acrylic acid, optimal conditions at 70 °C	Water absorption occurs through hydrogen bonding between hydroxy groups and water molecules, forming a gel-like structure that retains moisture	Phan et al. (2020), Alam et al. (2019)
Cellulose Hydrogel (cross-linked with borax)	~ 900	Derived from water hyacinth cellulose	Cross-linking creates a three-dimensional network that traps water molecules, leading to significant swelling	Tanpichai et al. (2022)
Bacterial Cellulose Composite SAP	~ 350	Enhanced by incorporating natural polysaccharides	Bacterial cellulose exhibits a high surface area and porosity, allowing for extensive water retention through capillary action and hydrogen bonding	Luo et al. (2019)
Cellulose Nanocrystal-based SAP	~ 817 g/g	Combined with chitosan; enhanced mechanical properties	The high aspect ratio of nanocrystals increases surface interaction with water, leading to enhanced absorption through capillary forces	Ismaelilmoghadam et al. (2023)

Table 4 Comparison of the different polymerization techniques

Polymerization Type	Process	Characteristics	References
Solution polymerization	The monomer is dissolved in a non-reactive solvent with an initiator or catalyst. Polymer formed is soluble in the solvent	Stability, affordability, good heat transfer. Easy to control molecular weight, Suitable for high-viscosity polymers, Solvent needs to be removed, Limited to monomers soluble in the solvent	Odian (2004), Wan Ibrahim and Mujiaba (2012), Ma et al. (2015), Ma and Wen (2020)
Graft polymerization	Monomer grafted onto a polymer backbone, forming branches. Initiators used to generate free radicals	Improves polymer properties, Versatile technique, Difficult to control grafting density, Requires specific monomers and polymers	Savina et al. (2006), Sherazi (2016), Cheng et al. (2019), Purohit et al. (2023)
Inverse-phase suspension polymerization	Water-in-oil emulsion, with water as the dispersed phase and monomer dissolved in oil	Complex, higher costs, and unstable; Particle products, Suitable for water-soluble monomers, requires specific monomers and solvents, Surfactants needed	Amjad et al. (2022), Pladis and Kiparissides (2014)
Radiation polymerization	Polymerization initiated by high-energy radiation like gamma rays, electron beams, or UV light	Fast heat, high efficiency, and clean. Initiator not needed, Suitable for sterilization	Bardajee et al. (2009), Adrovic (2012b), Ghobashy (2018), Bardajee et al. (2009)

Cellulose nanocrystals

Nanocrystals in the hydrogel structure augment the water absorption. Cellulose nanocrystals were created from carboxymethylcellulose sodium salt and hydroxyethyl cellulose and crosslinked with citric acid. The superabsorbent hydrogel showed a 600% swelling rate. The SAP is useful for water resource utilization in agriculture (Das et al. 2021).

A coating solution for NPK fertilizers was manufactured from polyvinyl alcohol (PVA) nanocomposites packed with cellulose nanocrystals (CNC). The coating significantly prolonged nutrient release and enhanced the water-holding capacity of the soil. CNC concentrations affect the water-holding capacity of the hydrogel (Kassem et al. 2021).

Bio-based superabsorbent aerogels manufactured from cellulose nanofibers (CNFs) show enhanced water absorption capacity and are potential substitutes for customary superabsorbent polymers in diapers (Patiño-Masó et al. 2019). Table 3 summarizes the comparison of different cellulose-based SAPs.

Characteristics of cellulose-based SAPs

Cellulose-based superabsorbent polymers (SAPs) are gaining attention due to their environmentally friendly properties and potential applications in various fields, including agriculture and medicine. Understanding their key characteristics is essential for evaluating their performance and suitability for specific applications.

Water absorption rate and retention capacity

The water absorption rate and retention capacity are critical properties of cellulose-based SAPs that determine their effectiveness in various applications, particularly in agriculture and biomedical fields. Cellulose-based SAPs have demonstrated impressive water absorption capabilities. For instance, a novel cellulose-based hydrogel achieved a water retention value (WRV) of 725 g of distilled water (d-water) per gram of gel, significantly surpassing the typical WRV of conventional cellulose-based materials, which ranges from 10 to 100 g/g. This performance is comparable to synthetic superabsorbent polymers, which can reach WRVs of up to 1000 g/g in d-water. In saline conditions (0.9% NaCl), the same hydrogel

maintained a WRV of 118 g of saline water (s-water) per gram of gel, more than double that of commercial synthetic gels, which usually range from 40 to 50 g/g (Alam et al. 2019).

Moreover, cellulose-based SAPs exhibit excellent re-swelling properties, losing only about 5–10% of their absorption capabilities after multiple uses. This characteristic is essential for applications requiring repeated cycles of hydration and dehydration, such as in agricultural soil amendments or hygiene products. Cellulose cross-linked with succinic anhydride results in a gel with a water absorption rate of 400g/g (Yoshimura et al. 2006). Free swelling capacity is a good parameter for the evaluation of SAP efficiency (Tomar et al. 2007). A cellulose-derived hydrogel shows the capacity to absorb 18 times its weight in a low-salt solution. The performance of this hydrogel is 1.8 times lower than that of synthetic gel. However, cellulose-based gels are biodegradable and environmentally friendly. A cellulose-chitosan cross-linked hydrogel absorbed up to 610 g d-water/g gel (Arredondo et al. 2023). However, the water absorption capacity of cellulose-based superabsorbent materials is less than that of synthetic hydrogels (Alam and Christopher 2018).

The ability of these hydrogels to absorb and retain water is influenced by their chemical structure and cross-linking density. Increased carboxyl group content within the polymer enhances its ionic charge, which correlates with improved water absorption capacity. The cross-linking process also plays a pivotal role; optimal cross-linking can create a network that allows for significant swelling while maintaining structural integrity. However, excessive cross-linking can limit water absorption due to a denser structure that restricts water flow into the gel.

Biodegradability

One of the most significant advantages of cellulose-based SAPs is their biodegradability and biocompatibility. Cellulose-based materials are inherently biodegradable, breaking down into non-toxic byproducts in natural environments. Cellulose, being a natural polymer, is readily degraded by microorganisms such as bacteria and fungi. Microorganisms produce cellulase enzymes that break the β -1,4-glycosidic linkages in cellulose (Moon et al. 2011). The hydrolyzed cellulose is broken down into glucose, which

microorganisms metabolize into water, carbon dioxide, and biomass. Cellulose derivatives with lower DS (fewer substituted hydroxy groups) degrade faster due to better enzyme accessibility (Habibi et al. 2010). These SAPs exhibit minimal toxicity to biological systems, making them suitable for applications in medical fields, such as wound dressings or drug delivery systems. This characteristic makes them preferable over synthetic alternatives, which can persist in ecosystems. In contrast, the synthetic SAPs degrade into small oligomers or microplastics. Both are toxic to organisms and the environment (Filicotto and Rothenberg 2021; Lim 2021). The cellulose-based SAP degrades through aerobic decomposition by the activity of microorganisms. These hydrogels are 58.1% more decomposable in contrast to fossil fuel-based SAP (Arredondo et al. 2023).

Mechanical properties

Cellulose-based superabsorbent polymers (SAPs) exhibit a range of mechanical properties that are crucial for their performance in various applications, particularly in agriculture and construction.

Tensile strength is one of the key mechanical properties, referring to the maximum amount of tensile stress a material can withstand before failure. The tensile strength of cellulose-based SAPs is significantly influenced by their composition and structure. For instance, incorporating cellulose nanofibers can enhance tensile strength due to their high aspect ratio and large surface area, which improve interfacial bonding within the polymer matrix (Susmanto P 2023). Additionally, cross-linking agents play a critical role in enhancing tensile strength by creating a network structure that increases the resistance of the polymer to deformation under stress (Hubbe 2013).

The elastic modulus indicates a material's stiffness or resistance to elastic deformation when stress is applied. The elastic modulus of cellulose-based SAPs can be influenced by the type and concentration of cellulose used; higher cellulose content typically leads to increased stiffness due to stronger intermolecular interactions (Hubbe 2013). Environmental factors, such as moisture presence, can alter the elastic modulus as well; when these materials absorb water, they may become softer and less stiff, which is an important consideration for applications requiring

consistent mechanical properties under varying humidity conditions (Zheng et al. 2021).

Compressive strength is another important property that measures a material's ability to withstand axial loads without failure. While cellulose-based SAPs generally exhibit lower compressive strength compared to synthetic polymers, modifications such as blending with other polymers or incorporating fillers can significantly enhance this property (Omidian et al. 2024). For example, adding materials like kaolin or other nanocomposites has been shown to improve both compressive and flexural strengths. The microstructure of cellulose-based SAPs, characterized by pore size and distribution, also affects compressive strength; a well-structured porous network can distribute loads more evenly, enhancing durability under compression (Zheng et al. 2021).

Stability and durability are essential for the long-term performance of cellulose-based SAPs. These materials generally exhibit good thermal stability; however, excessive heat can lead to degradation. Modifications with thermally stable additives can enhance this property (Omidian et al. 2024). Furthermore, these materials should maintain their mechanical properties when exposed to various environmental conditions (e.g., moisture and temperature fluctuations). Research indicates that specific formulations can improve resistance to degradation from environmental factors.

Swelling mechanism of SAPs

Several mechanisms contribute to the remarkable swelling capacity of SAPs. One key factor is the inherent hydrophilicity of the polymer backbone, often containing carboxylate acid groups. When exposed to water, these hydrophilic groups readily form hydrogen bonds with water molecules, a process known as hydration. This interaction raises the system's entropy, driving the spontaneous attraction of water molecules towards the polymer chains.

Beyond simple mixing, partially neutralized SAPs exhibit another potent swelling mechanism. Within the gel network, sodium ions associated with the neutralized carboxylate groups become mobile due to the weakened electrostatic interactions induced by water molecules (Yu et al. 2019a). This allows the gel to act as a semi-permeable membrane, creating an osmotic pressure gradient between the gel inside

and the surrounding fluid. Studies have shown that this osmotic pressure difference, rather than mixing alone, constitutes the primary driving force for the significant swelling observed in SAP gels (Yu et al. 2019a). Figure 7 shows the schematic structure of a SAP molecule.

When a polymer containing sodium interacts with water, the sodium ions spread uniformly between the polymer network and the water, causing the polymer to absorb and hold the water to form a hydrogel. Hence some sodium atoms desire to exit the network and move to water. When these sodium atoms move, they are substituted with water molecules. Water swells the polymer network, balancing the salt concentrations between the polymer and water. The crosslinks between the chains protect them from separating or dissolving in water (Yamuna and Kandhavadi 2023).

Factors affecting swelling of SAPs

The cross-linking agent determines the degree of cross-linking and hence the three-dimensional network structure of the SAP. The cross-linking agent affects the elasticity, viscosity, solubility, melting point, strength, and toughness of the SAP (Maitra 2014). The gel point times of SAP vary with the kind and quantity of cross-linking agents. This results in a pore-making effect (Kabiri et al. 2003). The ion concentration affects free water absorption from the environment through the osmotic pressure difference between SAP and the environment. Moreover, the inhibition of ion group dissociation and the charge screening effect also affect water absorption (Zhang et al. 2021). The difference in the concentration gradient between the SAP and the environment causes the diffusion of water from the environment into the SAP. The increase in ion concentration of external solutions decreases the water absorbance. SAP absorbs more water from monovalent cationic saline solutions as compared to multivalent cationic saline solutions (Xu et al. 2014). The pH of the SAP environment also influences the water absorption potential of SAPs (Fateme Soleimani 2012). Studies revealed that SAP absorbs maximum water at pH solution 8 up to 382 g/g (Adair et al. 2017). At low pH, the protonation of the polymer decreases, the swelling capacity of SAP and the polymer behaves hydrophobic. While at higher pH, the swelling capacity of SAP declines

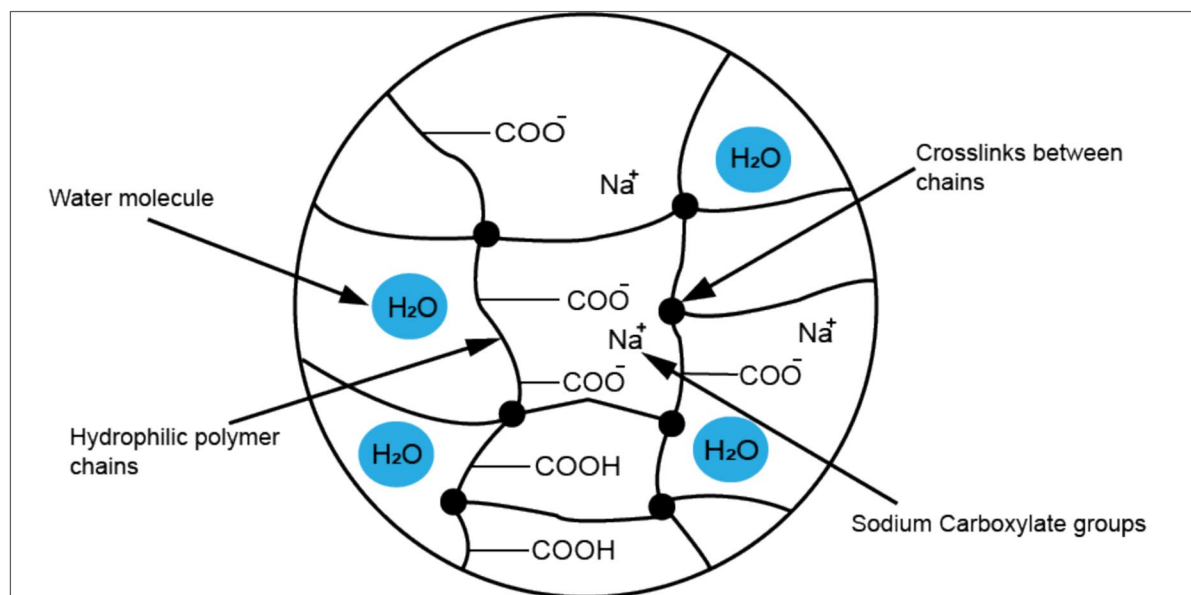


Fig. 7 Schematic structure of a SAP gel particle (Yu et al. 2019a, b)

due to the “charge screening effect” of surplus Na^+ in the solution. Na^+ ions shield the anions in polymer and inhibit effective anion-anion repulsion. Maximum water is absorbed when the groups in the polymer are ionized. At $\text{pH}=8$, some of the carboxylic acid groups are ionized and the electrostatic repulsion between COO^- groups causes an enhancement of the swelling capacity (Wu et al. 2012). The repulsion between ionized carboxylic groups causes polymer chain expansion and hence increased pore network (Rosa and Casquilho 2012). Cellulose-based SAP absorbed water amplified by an approximate factor of five (from 475 to 2248 g/g) (Arredondo et al. 2023).

SAP preparation process

The preparation of superabsorbent polymers begins with the interaction of a monomeric compound possessing hydrophilic groups with an initiator, crosslinking agent, and sodium hydroxide (Chen et al. 2005; Gooch 2011). The monomer is then polymerized by the subsequent reaction of the monomer, which results in the sodium salt of the polymer. The hydrogel is then crosslinked with a crosslinking agent to form the superabsorbent polymer, which can then be dried and powdered. The polymerization reaction initiated by free radicals forms a three-dimensional

network, giving superabsorbent polymers (SAPs) their unique ability to absorb and retain vast quantities of water or aqueous solutions relative to their mass. However, as the cross-linking density increases, the absorbency decreases (Omidian et al. 1994).

Methods of cellulose-based SAP synthesis

Solution polymerization

Most cellulose-based SAPs are prepared through the solution polymerization method. Solution polymerization is preferred because it allows for greater control over polymerization temperatures, is less expensive, and is more convenient. This technique is used to create the vast majority of cellulose-based superabsorbent hydrogels. When the initiator is activated by temperature or radiation, the polymerization process starts (Ma et al. 2015). To increase heat transfer during solution polymerization, a solvent is added to the reactor (Oadian 2004). At the beginning of the process, the initiator and solvent are introduced into the reactor. The initiator (UV radiation or a redox initiator system) will start the polymerization process by decomposing to produce primary free radicals. The free radicals then react with the monomer

usually acrylic acid or acrylamide, which is present in the reactor as a homogeneous solution. This phase continues until the reaction reaches completion. The efficiency of the initiator determines how effectively the initiator molecules decompose (Wan Ibrahim and Mujtaba 2012). Figure 8 shows the flowsheet diagram of the solution polymerization process.

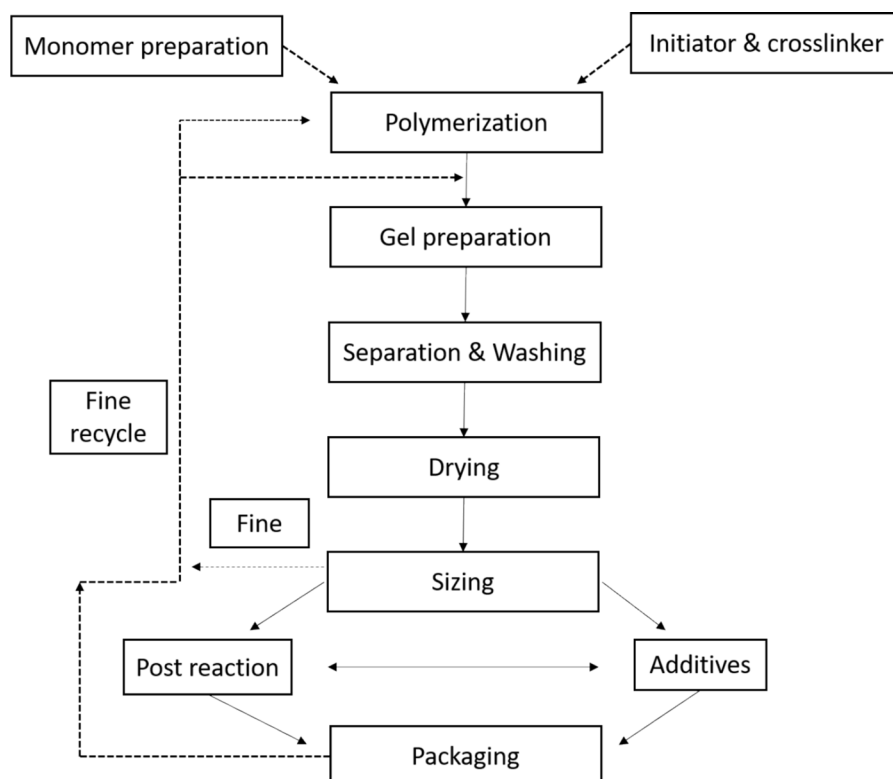
Usually, solvents like benzyl alcohol, ethanol, water, and their mixtures are used in solution polymerization (Mathur et al. 1996; Memic et al. 2016). Careful solvent selection is crucial to minimize chain transfer reactions hindering polymer growth (Su 2013). Both the monomer and initiator, as well as the resulting polymer, must be able to dissolve in the chosen solvent or mixture. This technique involves mixing ionic or neutral monomers with a multifunctional crosslinker. After that, polymerization is started thermally, by UV radiation, or using a redox initiator system. Following hydrogel production, the solvent can be eliminated by an additional hydration process by swelling the hydrogels in deionized water (Argenta et al. 2019). Kim et al. synthesized a cellulose-based SAP by solution polymerization method using itaconic acid and vinyl sulfonic acid and then treated it

with nanocellulose. The superabsorbent polymer with nanocellulose demonstrated outstanding absorption performance. The optimal nanocellulose content was approximately 2 wt%, which maximized the SAP's water absorption capacity. The addition of nanocellulose enhanced the hydrophilicity of the SAP, thereby improving its absorption efficiency (Kim et al. 2020). The advantage of using the solution polymerization method is that the reactants are completely dissolved in a solvent, ensuring a uniform and controlled reaction system. This leads to more consistent and predictable product properties. The temperature and molecular weight of the polymer can also be controlled easily and the viscosity of the system is also lower compared to bulk or suspension polymerization (Ma and Wen 2020).

Graft copolymerization

A common type of solution polymerization is graft copolymerization. Graft polymers are also known as graft copolymers because they contain at least two different types of monomer units, such as the grafted side chains, which differ structurally from the main

Fig. 8 Solution Polymerization technique (DEVI and GABA 2019)



chains. To initiate graft copolymerization, a variety of methods can be used, including chemical treatment, photochemical treatment, ionizing radiation (e.g., gamma radiation, electron beam radiation), photo-irradiation, plasma-induced techniques, and enzymatic grafting. Grafted polymers can be extremely useful because they can be adjusted to the needs of individual applications by selecting the appropriate backbone monomers. (Sherazi 2016). Monomers are usually grafted on naturally occurring polymers e.g. cellulose (Cheng et al. 2019), pectin, starch, Chitin (Vieira et al. 2014) etc. Figure 9 shows the schematic reaction of graft copolymerization of cellulose.

Enfa et al. successfully synthesized a new superabsorbent polymer composite by grafting acrylic acid (AA) and acrylamide (AM) onto α -cellulose chains, with magnesium zeolite as an inorganic filler. Compared to poly(AA-co-AM), the SAP composites showed a significant increase in water absorbency—93.88% in distilled and 89.58% in a 0.9 wt.% NaCl solution. Additionally, the water retention capacity improved by 71.79%. The use of magnesium zeolite also increased the thermal stability of the SAP composites (Fu et al. 2022). Suo et al. utilized free-radical grafting solution polymerization of acrylic acid and acrylamide monomers onto carboxymethyl cellulose (CMC) to create a new cellulose-based superabsorbent polymer, carboxymethyl cellulose-graft-poly(acrylic acid-co-acrylamide). They used N,N'-methylenebisacrylamide (MBAm) as a crosslinker and a redox couple of potassium persulfate and sodium metabisulfite as initiators. The optimized product exhibited a maximum water absorbency of 920 g/g in distilled water and 85 g/g in the aqueous NaCl solution. Additionally, the superabsorbent demonstrated good water retention and salt resistance

(Suo et al. 2007b). Luo et al. developed another cellulose-based superabsorbent by grafting acrylic acid onto bacterial cellulose with cyclohexane as a solvent using MBAm as a crosslinker and ammonium persulfate as an initiator. The BC was stirred in cyclohexane under an inert nitrogen atmosphere for 30 min to remove trapped oxygen and ensure a uniform reaction environment at 55–75°C. The acrylic acid was partially neutralized (70–90%) before being introduced into the reaction system. The composite achieved a maximum water absorbency of 322 ± 23 g/g in distilled water. However, the water absorbency was significantly lower in various salt solutions, and the absorption capacity decreased as the salt concentration increased (Luo et al. 2018). Grafting allows for the introduction of specific functional groups or monomers onto the cellulose backbone, tailoring the SAP properties to specific needs (Thakur and Thakur 2014). By grafting hydrophilic monomers, the water absorption capacity of cellulose can be significantly enhanced, making it suitable for applications like diapers, sanitary products, and agriculture. An important advantage of grafting is that it can enhance the compatibility of cellulose with other materials, facilitating the creation of composite materials with desired properties. However, graft copolymerization is often more complex than other polymerization techniques, such as free radical polymerization and addition polymerization, as it requires careful control of reaction parameters such as temperature and monomer and initiator concentration to ensure desired polymer properties (Kumar et al. 2017). The concentration of monomers, such as acrylic acid in graft copolymerization, significantly affects the rate of grafting and the properties of the final product. Higher concentrations can increase the likelihood of successful grafting but

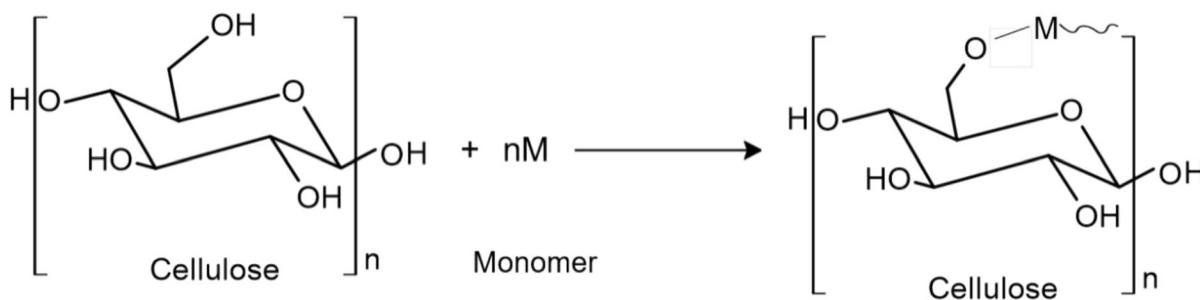


Fig. 9 Schematic representation of general graft copolymerization of cellulose

may also lead to increased viscosity, which can hinder the movement of radicals and reduce efficiency (Purohit et al. 2023). The amount and type of initiator (e.g., ammonium persulfate) influence the generation of free radicals, which are essential for initiating the polymerization process. The efficiency of the initiator directly impacts the rate of polymerization and the molecular weight of the resulting copolymer (Savina et al. 2006).

Inverse-phase suspension polymerization

In inverse emulsion polymerization, a hydrophilic monomer or its aqueous solution is emulsified within a continuous non-polar phase with the use of surfactants as shown in Fig. 4. The hydrophilic monomer (or its solution) is dispersed into tiny droplets within an oily phase. These droplets are stabilized by an emulsifier system with a low hydro-lipid balance (HLB). The polymerization reaction occurs within these droplets. As the polymerization process concludes, these droplets turn into solid particles (Amjad et al. 2022).

Inverse-phase suspension polymerization differs from precipitation polymerization in that, during inverse-phase suspension polymerization, neither the monomer nor the resulting polymer is soluble in the continuous phase. However, they are soluble in each other, allowing for effective polymerization to occur. The rate of polymerization is typically higher than in solution, resulting in a higher weight-average molecular mass for the resulting polymer. In the fully transformed suspension, the polymer concentration is usually between 30 and 50%. The viscosity of the suspension, which is influenced by the dispersion medium, remains virtually unchanged throughout the reaction, simplifying temperature regulation (Pladis and Kiparissides 2014). Microparticles or spherical hydrogels of the size 1 μm to 1 mm can be prepared by this technique. Klinpituksa et al. reported a hybrid sodium carboxymethyl cellulose and polyacrylic acid SAP using the inverse phase polymerization method as shown in (Fig. 5). The absorption capacity was 544.95 g/g in distilled water and about 44.0 g/g in the NaCl solution.

Guan et al. synthesized a cellulose-based SAP using the inverse-phase polymerization technique, by mixing cellulose, 100 g of cyclohexane, 17.3 g of acrylic acid, and 2.98 g of acrylamide at 45 °C for

30 min under nitrogen. After neutralizing the acrylic acid with sodium hydroxide, they added 0.08 g of potassium persulfate (KPS) as the initiator and 0.08 g of N,N'-methylenebisacrylamide (MBA) as the cross-linker, then raised the temperature to 75 °C to form a polymer gel. The gel was washed with ethanol and dried in a vacuum at 100 °C for 10 h to obtain a white powdered SAP. Through water and saltwater absorption studies, they investigated the parameters that affect water absorption, including temperature, oil-to-water ratios, neutralization degrees, monomer doses, and cross-linking agents. A cellulose monomer ratio of 1:10, a neutralization degree of 75%, an oil-to-water ratio of 3:1, and a reaction temperature of 70 °C were found to be the ideal reaction parameters. The polymer exhibited absorption rates of 859 g/g for water and 72.48 g/g for saltwater under these conditions (Guan et al. 2017). While this process can produce desired particle sizes, removing organic solvents such as toluene and n-hexane is a challenging procedure. Inverse-phase suspension polymerization is usually less preferred than aqueous solution polymerization due to its high cost and complexity hence, In the last five years, inverse-phase suspension polymerization has become less common.

Radiation polymerization

Radiation polymerization is a promising technique for producing high-quality, safe, efficient, and economical polymers (Ghobashy 2018). Various types of radiation sources can be used in the polymerization processes facilitated by radiation techniques. Ionizing radiations, such as gamma rays, accelerated electrons, and ion beams, cause polymerization reactions by forming highly reactive intermediates, known as energetic radicals (Ashfaq et al. 2020). These radicals, produced by ionizing radiation, interact with molecules, causing the ejection of electrons from a specific orbit, leading to the ionization of the atom. Other molecules can further separate these electrons when their orbits have the potential to overlap and create new bonds (Böhler et al. 2013). When the transferred energy is greater than the energy of the particular orbital electrons, this reaction occurs. Electrons become excited when there is not enough energy for ionization, which causes them to gain energy and shift to a higher energy level. Ionization is the major process since most molecules have an ionizing

potential of between 7 and 15 eV while ionizing irradiators have energies ranging from 1 to 100 MeV. Through energy transmission, energetic radicals with ionized electrons can excite another molecule (Gross 2017). Different reactions can be triggered by the ionized species linked to the energetic radicals generated by irradiation. This property in a polymerization reaction makes the procedure environmentally benign. Ionizing radiation causes covalent crosslinking of polymers at ambient temperature. Temperature and environmental factors, in addition to the radiation dose, determine the degree of crosslinking (Adrovic 2012a). Two molecules containing active radicals can unite head-to-head, head-to-tail, or tail-to-tail to form a crosslink. The combination of radicals is more common in an amorphous region than in a crystalline region. (Arutchelvi et al. 2008). An increase in crystallinity often indicates polymer crosslinking, while an increase in amorphousness indicates polymer degradation, also known as chain scission (Ghobashy and Abdeen 2016a, b). The type of polymer determines whether degradation or crosslinking takes place. It is possible to make radiation crosslinking dominant by substituting the oxidation gas (O_2) with an inert gas (N_2). The amount of polymerization and/or chain degradation can be regulated by the fraction of dissolved oxygen. Competition between the formation and breakdown of covalent bonds occurs during these processes. When it comes to vinyl monomers or synthetic polymers, the creation of new covalent bonds occurs more frequently (Ghobashy 2018). Figure 10 represents the mechanism of free radical polymerization of a vinyl monomer from cellulose (Figs. 11, 12, 13).

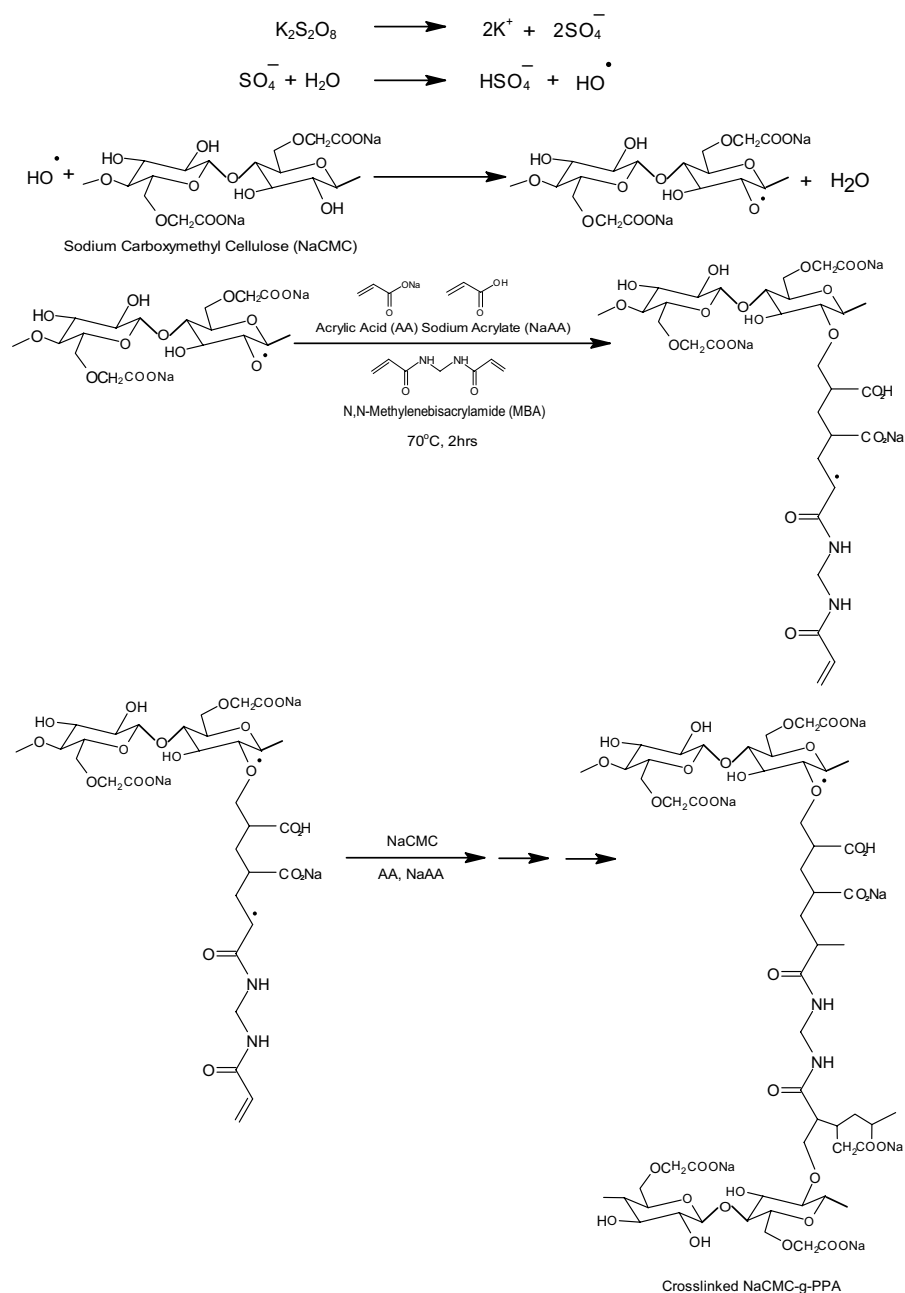
Ibrahim et al. researched to investigate the creation of superabsorbent hydrogels based on acrylamide monomer and crosslinked carboxymethyl cellulose polymer utilizing electron-beam irradiation. Polymer/monomer mixtures (2.5 g) of CMC and AAm was dissolved in water at room temperature. N,N-Methylene bisacrylamide (0.1% w/w) was added as a crosslinker, and the solutions were irradiated with a 1.5 MeV, 25 kW electron-beam accelerator at doses of 20–50 kGy, with a conveyor speed of 2.7 m/min. When pure water was used instead of salt solutions, the produced hydrogels swelled more, peaking at pH 7. In addition, the swelling was exacerbated by raising the temperature to 50 °C. The water-retention capabilities of these superabsorbent hydrogels were

the main focus of research on their possible application in rice cultivation (Ibrahim et al. 2007). Fekete et al. used gamma irradiation-initiated crosslinking to report superabsorbent hydrogels from aqueous solutions of four cellulose derivatives (Carboxymethyl cellulose (CMC), Methylcellulose (MC), Hydroxyethyl cellulose (HEC), and Hydroxypropyl cellulose (HPC)). Using MBAm as a crosslinking agent improved the gel fraction, increased crosslink density, and decreased water absorption. The maximum crosslinking efficiency was obtained at a polymer concentration of 1 w/w% MBAm. With MBAm, better water uptake was obtained under milder conditions without lowering the gel percentage (Fekete et al. 2016). Fekete et al. used gamma irradiation in a different investigation to investigate the impact of starch addition on the characteristics of CMC-based hydrogels (Fekete et al. 2017). Raafat et al. reported SAP for use in agriculture by crosslinking polyvinylpyrrolidone (PVP) with gamma irradiation and employing CMC. Recently, research has been done to examine how radiation exposure and montmorillonite concentration affect the physical characteristics. Salmawi et al. used gamma irradiation to create Superabsorbent hydrogel by combining carboxymethyl cellulose, acrylic acid, and clay montmorillonite (MMT) (Salmawi et al. 2018). Radiation-induced grafting is beneficial as it is quicker than chemical-induced grafting, thus saving time. The simplicity of the technique, absence of cross-linkers or initiators, lack of by-products, and competitively less operational costs make the radiation method an appropriate option for synthesizing Superabsorbent Polymers (Bardajee et al. 2009). Table 4 summarizes the comparison of polymerization techniques and their characteristics.

Physical methods

Physical synthesis methods involve the molecular assembly of polymers through cross-linking hydrogen or ionic bonds. Strong hydrogen bonds allow for the production of super-absorbent polymers at low temperatures. These connections could arise via freeze–thaw cycles (Guan et al.). Guan et al. prepared hydrogels from hemicelluloses, polyvinyl alcohol (PVA), and chitin nanowhiskers through multiple freeze/thaw cycles. They found that the increase in the number of cycles made the hydrogel stiffer, thermally stable, and more crystalline,

Fig. 10 Reaction mechanism for the synthesis of NaCMC-PAA SAP (Klin-pituksa and Kosaiyakanon 2017)



but after three cycles, the hydrogels' equilibrium swelling ratio dropped as a result of the lamellar structure forming (He et al. 2014). The thermal characteristics and hydrogen bonding of blends of poly(vinylphenol) (PVPh) and bisphenol A (phenoxy) polyhydroxyether (phenoxy) polymers were compared by Kuo et al. They discovered that PVPh and phenoxy tend to have hydrogen bonding that is

Fig. 11 Mechanism of free radical polymerization of a vinyl monomer from cellulose (Rodríguez-Fabià et al. 2022)

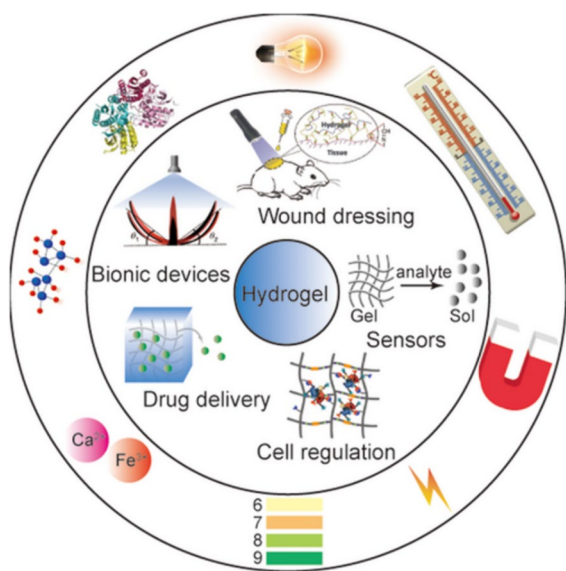
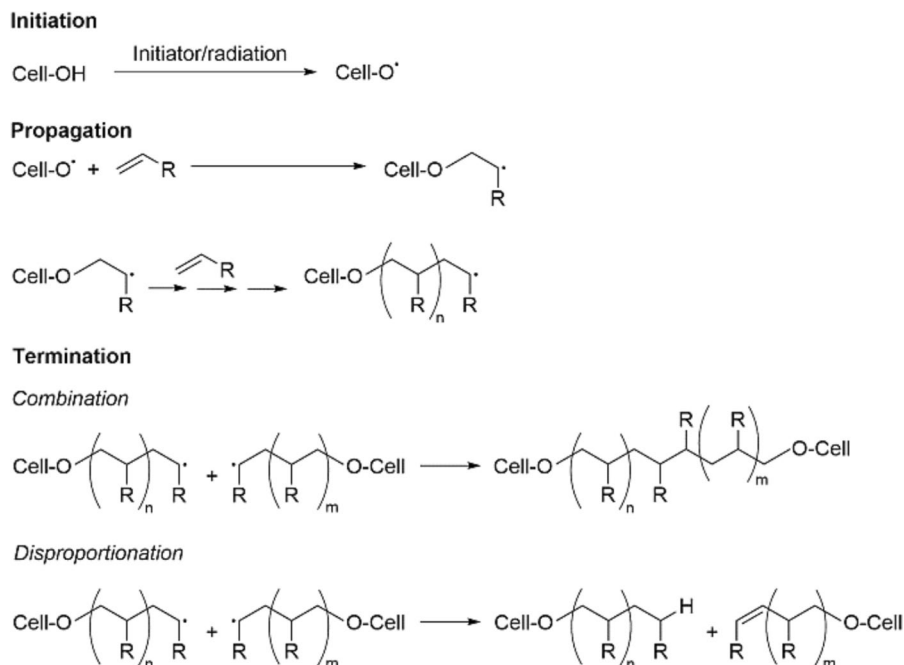


Fig. 12 Stimuli-responsive hydrogels and their applications (Li et al. 2022)

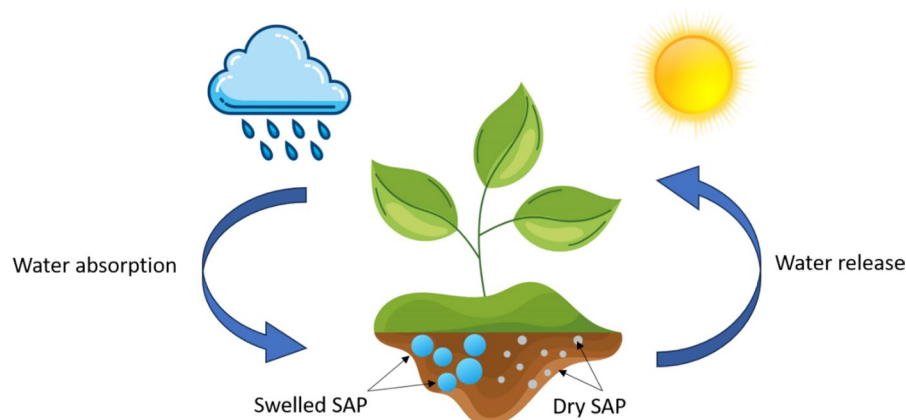
more favorable than their internal hydrogen bonding (Kuo et al. 2003).

Applications of cellulose-based SAPs

Controlled drug delivery systems

Cellulose ethers have been frequently used in the medical sector as inactive components in various medication device formulations (Baumgartner et al. 2002). They are used in tablets to enable a swelling-driven drug release when they come into contact with bodily fluids. Controlled drug release through oral administration typically relies on the significant pH changes that occur when passing from the gastrointestinal tract. Polyelectrolyte hydrogels based on cellulose, such as NaCMC, are especially well-suited for this purpose. For example, anionic hydrogels derived from CMC have recently been explored for drug delivery targeted to the colon (El-Hag Ali et al. 2008). The most recent advances in controlled release via a hydrogel matrix include delivering proteins, growth hormones, and genes to particular areas, a need driven by tissue engineering approaches. While hydrogel products for oral and transdermal distribution may be non-biodegradable, direct medication or protein delivery to diverse body sites necessitates hydrogel biodegradation to avoid foreign body reactions and eventual surgical removal. Injectable hydrogel compositions are especially promising and are

Fig. 13 Application of stimuli-responsive SAP in agriculture



currently being investigated. To avoid denaturing the loaded molecule, the crosslinking reaction must be performed under gentle circumstances, as should the microenvironment created by polymer degradation. Injectable medications based on HPMC have been developed to administer biomolecules and exogenous cells in vivo (El-Hag Ali et al. 2008). Different applications of stimuli-responsive hydrogels are shown in Table 4.

Wound healing

Hydrogels can create a moist environment that promotes rapid healing and is ideal for developing wound dressings. These can be sheets or amorphous hydrogels, which are typically physically crosslinked and decrease in viscosity as they absorb bodily fluids. They further offer non-adherent dressings that can be removed without damaging the wound. Numerous studies have been conducted to create cellulose-based materials for wound healing and dressing applications as described in Table 5. For instance, Agubata et al. reported a hydrogel to heal wounds by using HPMC, biodegradable microfibers from surgical sutures, and ofloxacin. These hydrogels led to significant collagen deposition 21 days post-wounding (Agubata et al. 2016). In 2016, Prabu et al. discovered that applying HPMC and HPC to sodium alginate hydrogel membranes loaded with gatifloxacin resulted in a prolonged release of the antibiotic, suggesting that these composites could be safe and effective materials for wound dressings (Prabu et al. 2016). Ogawa et al. (Ogawa et al. 2014) developed a minimal-substituted HPC hydrogel with less adhesive ability that

Table 5 Commercially available wound dressings containing CMC and NaCMC

Product name	Manufacturer	Main components
AQUACEL	ConvaTec	NaCMC
Hydrofera Blue	ConvaTec	NaCMC, blue dye
Kaltostat	ConvaTec	NaCMC, calcium alginate
Carboflex	ConvaTec	NaCMC, alginate
Purilon	Coloplast	NaCMC
Comfeel	Coloplast	Hydrocolloids (includes cellulose derivatives)
Medihoney	Derma Sciences	Honey, NaCMC
Hydrogel	Smith & Nephew	HPMC
Tegaderm Hydrogel	3 M	HPMC
Sofra-Tulle	Sofar	HPMC, paraffin

could maintain adequate moisture for wound healing. Rakhshaei et al. synthesized a nanocomposite hydrogel using CMC, mesoporous silica, and zinc oxide for wound dressing (Rakhshaei et al. 2017). In the same year, Park and his associates used a CMC hydrogel paste saturated with chestnut honey to treat diabetic ulcer wounds (Park et al. 2017). Furthermore, hydrogels with long-lasting antibacterial qualities will help patients whose wounds need to be dressed every day and will also help create new technologies for making scaffolds. Kondaveeti et al., in the meantime, created films using gentamicin, xyloglucan, and HPMC and examined their antibacterial effectiveness against *Escherichia coli* and *Staphylococcus aureus* (Kondaveeti et al. 2017).

Personal hygiene products

Superabsorbent hydrogels (SAPs) synthesized from acrylates are widely used in personal care products like diapers and adult incontinence products to absorb fluids, promoting skin health and comfort (Noonan et al. 2006). They help keep skin dry, prevent diaper rash, and control germ spread in group care settings. Disposable diapers containing SAPs are found to be more effective in minimizing the risk of gastrointestinal illnesses spread than cloth diapers. Unicharm of Japan used SAPs in the diaper sector for the first time in 1982, leading to a new era of high-performance, thinner diapers with improved retention performance (Tuteja and Vyas 2023). This reduced leakage and diaper rash, and also decreased the average weight by 50%, benefiting both the environment and packaging costs. However, the ecological impact of disposable diapers is significant. A child uses about six diapers a day, producing around 1,092 cubic meters of garbage a year. With approximately 50,000 diaper consumers per million population, a city of one million would need to dispose of 150 cubic meters of diapers daily (Mondal and Haque 2019). Efforts have been made for the recycling of disposable diapers and similar products. However, the process's complexity has led to the development of biodegradable diapers, which still contain the non-degradable acrylate-based SAP. A recent solution for SAP recycling is the use of totally biodegradable and cellulose-based hydrogels. These novel hydrogels, based on sodium carboxymethylcellulose (NaCMC) and HEC crosslinked with DVS, have comparable absorption capabilities, by inducing microporous structures (Sannino et al. 2004). The primary benefit of cellulose-based hydrogels compared to current SAPs is their biodegradability and eco-friendly properties. Recent advancements in these cellulose-based hydrogels include the development of an environmentally sustainable manufacturing method and the utilization of non-toxic agents for cross-linking. The technique of cellulose radiation crosslinking, which eliminates the need for additional chemicals, could be instrumental in creating new eco-friendly superabsorbents.

Agricultural applications

The use of superabsorbent hydrogels is gaining traction in agriculture, primarily due to the demand

for water conservation and efficient use of water resources in farming and gardening (Abd El-Rehim et al. 2004). Hydrogels, during their swelling process, transition from transparent to a rubber-like phase, capable of storing water under substantial compression. If there is a humidity gradient, the swelled hydrogel will progressively release water via a diffusion mechanism. Alongside water, other molecules incorporated into the polymer network can also be released in a controllable and sustained manner. When watered, the hydrogel absorbs the water and then releases it, maintaining the moisture in the soil over long periods (Supare and Mahanwar 2022) as shown in Fig. 1. This process results in significant water savings by preventing water loss due to evaporation and drainage and allows for the redistribution of available water resources.

Another advantage of using hydrogels is the impact of their swelling on the soil. The hydrogel granules, similar in size to substrate granules when dry, increase in size upon swelling, enhancing soil porosity and improving oxygenation to plant roots. This suggests that hydrogels with larger granules may yield better results than those with finer granules when properly mixed with the soil (Behera and Mahanwar 2020).

A lot of commercial products are acrylate-based and non-biodegradable, but cellulose-based hydrogels provide an environmentally friendly alternative. A new class of fully biodegradable microporous cellulose-based superabsorbent hydrogels has been developed recently. These hydrogels may absorb up to a thousand grams of water without releasing it during compression (Cannazza et al. 2014). They can be made in powder form or as a thick material with a well-defined shape that holds its shape even after swelling. Small molecules, such as nutrients, can be put into the material and released in controlled amounts. Studies have been done to investigate the use of SAPs as a water reservoir in agriculture, examining the hydrogel's absorption capacity at various ionic strengths of the swelling media in order to imitate the hydrogel-soil interaction as precisely as possible (Das et al. 2021).

Body water retainers

Hydrogels based on cellulose, because of their natural biocompatibility and versatile properties,

are attractive for various *in vivo* applications. One such application is their potential use as devices for extracting excessive amounts of water from the body, useful in treating conditions like diuretic-resistant edemas and renal failure (Ambrosio et al. 2011). These powdered hydrogels are meant to be ingested and, in the acidic environment of the stomach, absorb water as they move through the intestine, where the pH is approximately 6–7. This prevents swelling. The hydrogel subsequently exits the body through feces, carrying out its intended purpose without interfering with physiological processes (Sannino et al. 2003). Because of their sensitivity to pH, polyelectrolyte cellulose hydrogels based on NaCMC and HEC have been investigated for such applications. At neutral pH, these hydrogels have good swelling characteristics, while at acidic pHs, they have poor swelling ratios. Moreover, the NaCMC/HEC ratio, the quantity of crosslinker (divinylsulphone), and the addition of molecular spacers, like polyethylene glycol (PEG), to the polymer network can all be changed to optimize their water absorption capacity (Esposito et al. 2005). The studied hydrogel formulations showed good biocompatibility in both *in vitro* and *in vivo* studies, even with the usage of DVS as a crosslinker. When used with diuretic therapy, hydrogels have the potential to take the place of some medications and eliminate water from the body through the intestinal route as opposed to the systemic one.

Stomach bulking agents

Dietary supplements for obesity are thought to function either by directly suppressing appetite, as is the case with some natural fibers and herbal products that seem to absorb fluids and expand in the stomach, developing a feeling of satiety, or by binding fats, thereby reducing fat absorption, as seen with chitosan-based products (Pittler and Ernst 2004). Another approach that relies on natural fillers or bulking agents is very interesting since it can significantly reduce food intake by making less room in the stomach, which can avoid the necessity for intricate surgical procedures. Nevertheless, although the adverse effects of existing bulking agents—mostly gastrointestinal symptoms—have a well-documented history, there is presently no conclusive evidence of their effectiveness in promoting weight loss over the long term (Saper et al. 2004). Furthermore, it's crucial to

remember that certain fillers, like guar gum, have been known to cause blockages in the stomach, esophagus, or intestines. Thus, the creation of novel bulking agents that effectively encourage weight reduction is required. Since their swelling capacity can be precisely engineered by regulating their chemical composition and physical microstructure, as well as by adjusting the environmental conditions (such as pH, ionic strength, and temperature), cellulose super-absorbent hydrogels are particularly intriguing in this regard (Sannino et al. 2006a). The fundamental concept is that before each meal, an oral pill containing xerogel is eaten; the xerogel powder expands in the stomach. As a result, there is less room for food consumption, which makes one feel full. The body then eliminates the enlarged hydrogel through feces. In this case, it is anticipated that the hydrogel will travel through the digestive system and come into contact with the different pH conditions found in the stomach and intestine (Chen et al. 1999).

New cellulose-based hydrogels made by crosslinking aqueous mixtures of NaCMC and HEC show promise as dietary bulking agents, in addition to superporous acrylate-based hydrogels that expand very fast in aqueous solutions. These hydrogels have a great capacity for pH-sensitive water retention and are highly biocompatible with intestinal tissues (Sannino et al. 2006b). Although the NaCMC network's polyanionic nature allows for larger swelling capabilities at neutral pHs than at acidic pHs, the swelling ratio at acidic pHs may still be important for the hydrogel's application as a stomach filler. In particular, hydrogels based on cellulose that are made using safe crosslinking agents are quite desirable for this kind of use.

Water treatment

Cellulose-based hydrogels have emerged as excellent wastewater treatment materials, especially for contaminants like heavy metals and dyes. They are suitable for long-term environmental applications because of their biodegradability and strong water absorption capacity (Persano et al. 2024a). The adsorption mechanism of these SAPs is founded on the interaction of pollutants with the hydrogel matrix, which can be altered by the presence of modifiers, pH, and ionic strength. For example, adding modifiers like tannic acid can improve the adsorption capability of heavy

metals like Cu(II) and other pollutants. Studies show that these hydrogels may selectively absorb metals like copper and lead, making them useful for cleaning industrial effluents (Akter et al. 2021). They can efficiently lower the concentration of different dyes, which are common pollutants in textile and dyeing industries (Persano et al. 2024b). Godiya et al. created a novel bio-based bilateral hydrogel with polyacrylamide (PAM) and CMC for wastewater treatment. Lead Pb(II), cadmium Cd(II), and copper Cu(II) ions showed a substantial single-ion affinity for the resultant CMC/PAM composite hydrogel. Furthermore, employing this Cu NPs-loaded CMC/PAM hydrogel, the reduction of 4-nitrophenol to 4-aminophenol was accomplished successfully and with great efficiency, demonstrating the hydrogel's dual uses for catalysis and wastewater treatment (Godiya et al. 2019). Demetri et al. focused on the creation of a biodegradable superabsorbent hydrogel based on cellulose derivatives in order to optimize water resources in horticulture and agriculture as well as, more generally, to encourage a more prudent and effective approach to water usage. The outcomes showed that the substance allowed for effective water storage and continuous water release to the soil and plant roots (Demetri et al. 2013). The optimization of SAPs based on cellulose for improved wastewater treatment performance is still being investigated. Future research is required to ascertain the stability over the long run, regeneration capabilities, and economic feasibility of these materials in large-scale applications. Furthermore, merging these SAPs with other materials may result in hybrid systems that offer enhanced adsorption properties and broader pollutant removal capabilities (Akter et al. 2021). Figure 14 summarizes the major applications of cellulose-based superabsorbent polymers.

Research Progress and Challenges in Cellulose-based SAPs

Recently, cellulose-based SAPs have been synthesized by the use of enzyme-catalyzed reactions. This is an eco-friendly method that evades the use of unsafe chemicals, contrary to synthetic polymers. Enzymes are selective in nature and enhance the water absorption property of SAP. A novel approach uses a solvent-free synthesis method to synthesize cellulose-based SAP. This method is advantageous



Fig. 14 Applications of cellulose-based superabsorbent polymers

because of the elimination of toxic solvents. The synthesis is safer and eco-friendly and reduces chemical waste. Recently, biocompatible, low-toxicity cross-linkers have been explored. Enhanced safety parameters of these materials recommend their use in hygiene and medical products.

The development of novel cross-linking agents has led to cellulose-based SAPs with improved mechanical strength and enhanced absorbency. Cross-linking increases the polymer's ability to retain water and swell without losing structural integrity. Recent advancements have focused on. Photo-cross-linking is a breakthrough in the synthesis of SAPs. In this technique, light is used to cross-link cellulose fibers. Photo-cross-linking permits the fine-tuning of the network structure that results in SAPs with precise performance necessities. For example, quick water absorption or slow release in agricultural applications. Nanocellulose offers greater surface area, exceptional mechanical strength, and biodegradability. These properties show nanocellulose as an ideal material for creating superabsorbents with greater performance. Cellulose nanocrystals (CNCs) show the capability to form highly absorbent networks. CNCs improve the overall efficiency of the SAP, enabling them to absorb larger amounts of water with greater speed while still being biodegradable.

A major challenge during the industrial manufacturing of cellulose-derived SAP is its cost-effectiveness. Complex extraction and processing methods of cellulose require advanced technologies. This may encompass an elevation in costs. Other factors contributing to high costs include specific solvents, energy, and equipment. Cellulose is extracted from plants. Climate change, struggle with food production, and utilization of land affect the availability of raw materials. The availability of raw materials and sustainability is another challenge. Maintenance of product stability and quality control during its manufacture and storage is essential. Developing consistent practices for quality-ensured products becomes a challenge and may increase the cost. Industries have invested in the utilization of synthetic SAP. In the case where traditional materials have well-developed infrastructure and demonstrated track records, the shift from customary materials to cellulose-based alternatives is slow. The adaptation of cellulose-based sap in the industry is still comparatively innovative and requires education. Application of cellulose-based sap food, pharmaceuticals, or agriculture may require regulatory compliance for safety, environmental impact, and efficacy.

As cellulose is a renewable source and eco-friendly hence the cellulose-based SAP is sustainable as compared to synthetic polymers. Utilization of plant materials also decreases dependence on petroleum-based products. Another advantage of cellulose-based SAP is its biodegradable nature. In contrast, synthetic SAPs are nondegradable and impact the environment. As research into cellulose extraction and processing technologies advances, production methods are expected to become more efficient, reducing costs. New advancements in biotechnology, such as microbial fermentation processes, could make cellulose-based sap more affordable and accessible on a larger scale. With economies of scale and technological improvements, these costs can decrease significantly over time. Certain industries, like packaging and construction, are now looking for sustainable and biodegradable products. Funding bodies also support the research exploring innovative sustainable products. Moreover, a variety of innovative products can be prepared from cellulose-based SAP by incrementing strength, elasticity, or water absorption capacity according to the demand of the application. This can be achieved by utilizing advanced technologies

and innovative use of sustainable materials that can decrease the cost of the product.

Conclusion

Cellulose-based superabsorbent polymers and hydrogels have garnered significant attention in recent years due to their biocompatibility and biodegradability, which makes them valuable in a variety of applications, including biomedical engineering, food packaging, environmental remediation, biomedicine, agriculture, water treatment, and personal health care. However, the development and application of these materials face several challenges in enhancing their performance for specific applications. For instance, achieving the desired balance between swelling capacity and mechanical strength can be difficult. Their performance can degrade under extreme conditions or over prolonged use, which is particularly critical for applications like drug delivery systems or wound dressings where reliability is paramount. Moreover, Cellulose's absorption capacity is limited compared to synthetic SAPs, such as polyacrylate-based systems. Chemical modification is often necessary to increase cellulose's hydrophilicity, but this can compromise its natural, biodegradable nature. Advancements in nanotechnology, including cellulose nanofibers (CNFs) and cellulose nanocrystals (CNCs), can offer improved surface area, mechanical properties, and water absorption capabilities. Deeper analysis is required to understand the kinetics of swelling of cellulose-based superabsorbent hydrogels in various media, as well as the multidisciplinary preparation mechanism of these hydrogels.

Recently, the development of SAP from cellulose has been the focus of attention for researchers. Cellulose is abundant, sustainable, and biodegradable. Contrary to synthetic products, cellulose-based SAPs are biobased and eco-friendly. Techniques have been developed to explore several renewable sources for cellulose extraction. Enzymatic methods, photo hydrolysis, and solvent-free extraction have significantly increased extraction efficiency and reduced cost and environmental impact. Nanocellulose as nanofibrils or nanocrystals offers high surface area, strength, and biocompatibility. These advancements increased the range of applications of cellulose-based SAPs. Nanocellulose also improves the stability and

functionality of the SAPs. Crosslinking cellulose with different chemical agents crafts a network structure that can retain large amounts of water. This development enables the use of cellulose-based SAPs in wound dressings, drug delivery systems, and tissue engineering. Cellulose-based films and coatings have been developed with varying degrees of biodegradability. Such materials are frequently used in packing, food protection, and biodegradable substitutes for nonbiodegradable plastics. Regardless of the talented perspective of cellulose-based sap, the manufacturing cost is a major fence for large-scale applications. There is much more to explore for effective processing methods, enhanced enzyme systems, and innovations in chemical modifications to reduce the cost. While cellulose-based materials have shown excellent mechanical properties, such as high strength and flexibility, researchers are still improving their durability, water resistance, and thermal stability to expand their range of applications. In summary, cellulose-based sap is being investigated for applications, from biodegradable films and hydrogels to biofuels and waste valorization. Despite challenges such as cost and material performance, ongoing research continues to explore ways to enhance the properties and scalability of cellulose-based materials, positioning them as a promising alternative to synthetic polymers in the context of sustainability and environmental protection.

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